

# Telecommunication Systems I

## Exercise 2

Antonios Moutsopoulos AM:2021030024

April 27, 2026

Total number of hours: 2 hours

### Part A: Baseband PAM waveforms

#### A.1 SRRC Pulse and ESD

We constructed an SRRC pulse  $\phi(t)$  with  $T = 10^{-3}$ ,  $over = 10$ ,  $A = 4$ , and  $\alpha = 0.5$ . The Energy Spectral Density (ESD) was computed using the FFT.

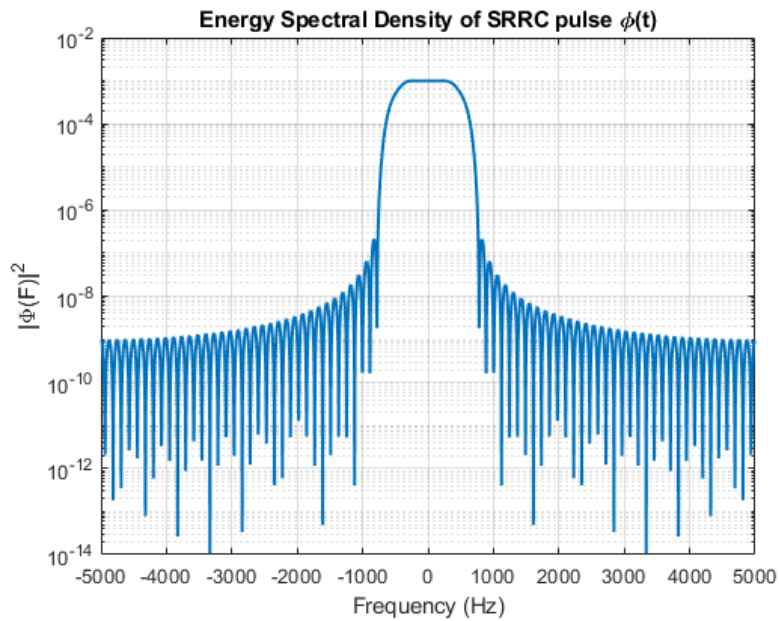


Figure 1: Energy Spectral Density  $|\Phi(F)|^2$  of the SRRC pulse.

#### A.3 2-PAM PSD Estimation

Using  $N = 100$  bits and  $K = 500$  realizations, we estimated the Power Spectral Density of the 2-PAM signal. The experimental result closely follows the theoretical curve  $S_X(F) = \frac{\sigma_X^2}{T} |\Phi(F)|^2$ , where  $\sigma_X^2 = 1$ .

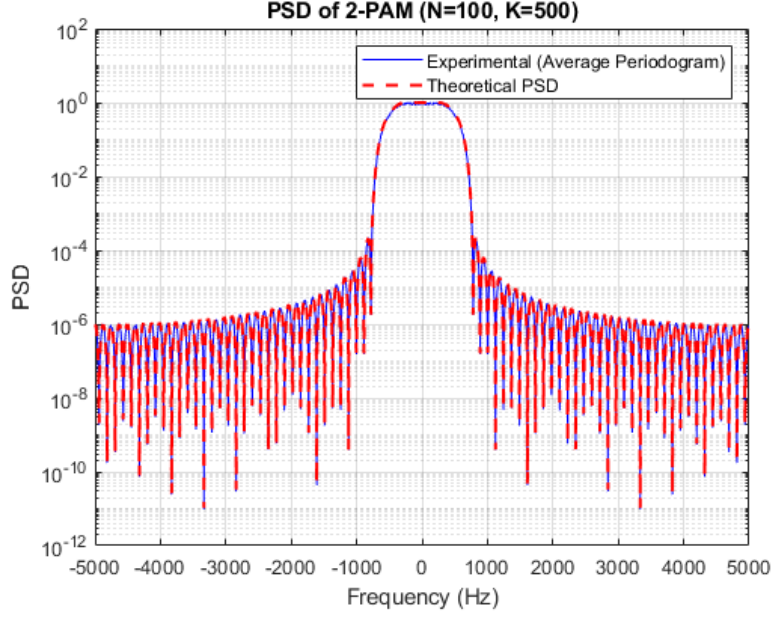


Figure 2: Experimental vs. Theoretical PSD for 2-PAM.

**Observation:** As  $K$  and  $N$  increase, the periodogram average becomes smoother and more accurate, providing a better approximation of the theoretical PSD.

#### A.4 4-PAM PSD Estimation

For 4-PAM, we mapped bit pairs to symbols  $\{+3, +1, -1, -3\}$ . The variance is  $\sigma_X^2 = 5$ .

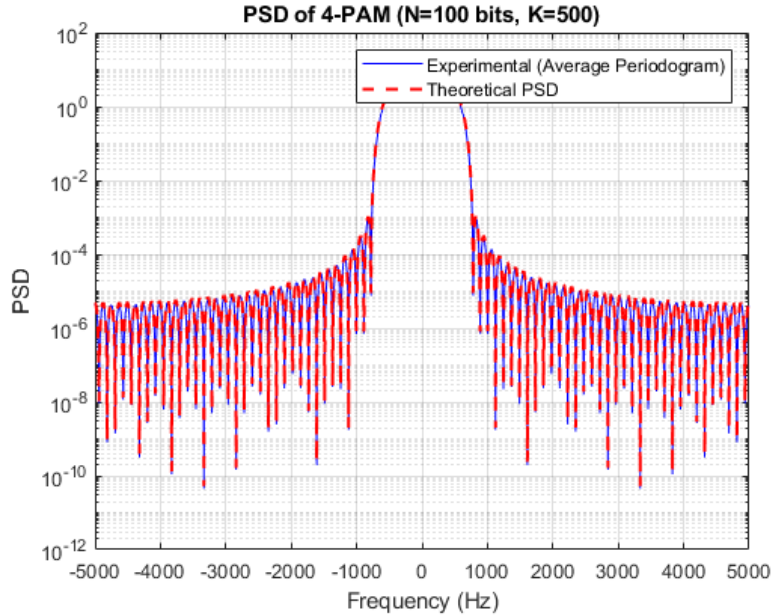


Figure 3: Experimental vs. Theoretical PSD for 4-PAM.

**Comparison:** The bandwidth of 4-PAM is identical to 2-PAM because they use the same pulse  $\phi(t)$ . However, the magnitude of the PSD is 5 times higher due to the increased variance of the symbols.

### A.5 Effect of Symbol Period $T' = 2T$

By doubling the symbol period  $T$ , the bandwidth of the signal is halved. This is consistent with the inverse relationship between time and frequency scaling.

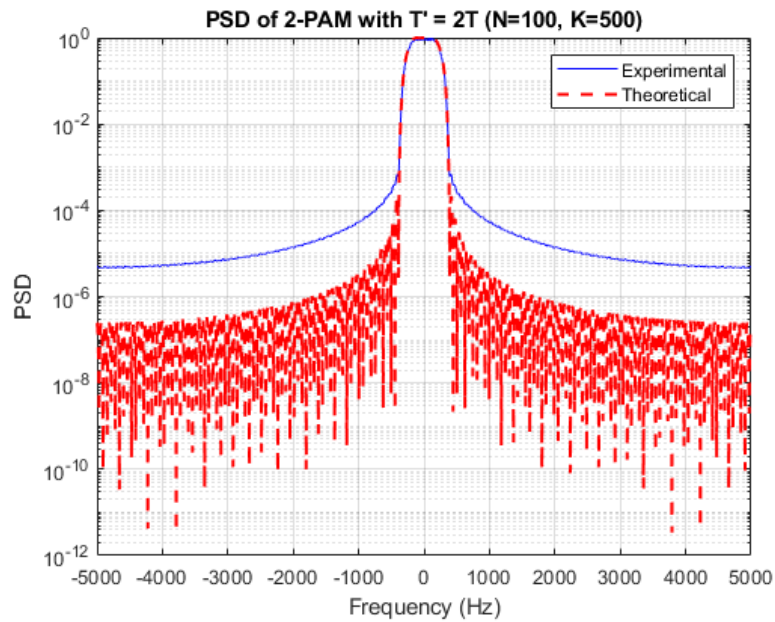


Figure 4: PSD with  $T' = 2T$  showing narrower bandwidth.

### A.6 Design Decisions

- To transmit data as fast as possible in the same bandwidth, we choose **4-PAM** because it carries 2 bits/symbol compared to 1 bit/symbol for 2-PAM.
- If bandwidth is expensive, we choose  $T' = 2T$  (or larger), as it reduces the spectral footprint.

## Part B: Stochastic Processes

Let  $Y(t) = X \cos(2\pi F_0 t + \theta)$  with  $X \sim N(0, 1)$  and  $\theta \sim U[0, 2\pi]$ .

### B.1 Realizations

Five realizations of the process are shown below.

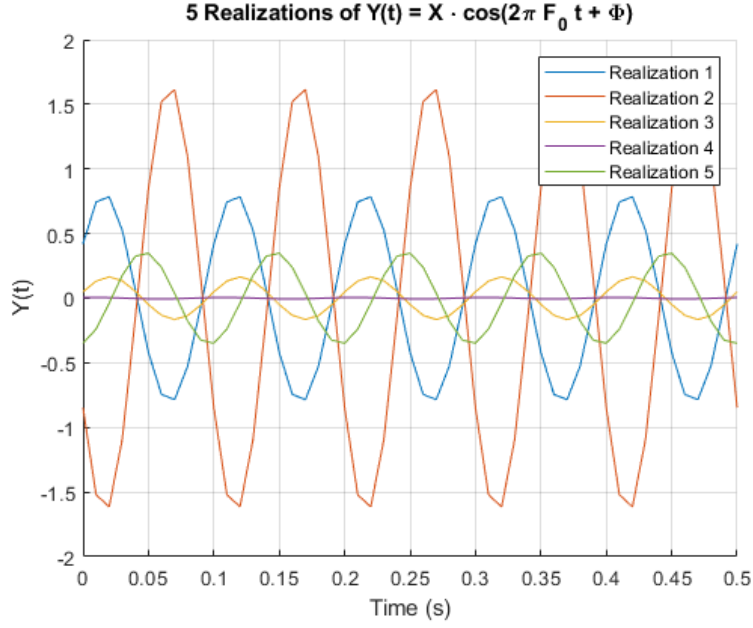


Figure 5: 5 realizations of the stochastic process  $Y(t)$ .

## B.2 Mean and Autocorrelation

**Mean:**

$$E[Y(t)] = E[X \cos(2\pi F_0 t + \Phi)] = E[X]E[\cos(2\pi F_0 t + \Phi)] = 0 \cdot E[\dots] = 0$$

**Autocorrelation:**

$$R_{YY}(t + \tau, t) = E[Y(t + \tau)Y(t)] = E[X^2 \cos(2\pi F_0(t + \tau) + \Phi) \cos(2\pi F_0 t + \Phi)]$$

Using the identity  $\cos A \cos B = \frac{1}{2}[\cos(A - B) + \cos(A + B)]$ :

$$R_{YY}(\tau, t) = \frac{1}{2}E[X^2]\{\cos(2\pi F_0 \tau) + E[\cos(2\pi F_0(2t + \tau) + 2\Phi)]\}$$

Since  $E[X^2] = 1$  and  $E[\cos(\dots + 2\Phi)] = 0$ :

$$R_{YY}(\tau) = \frac{1}{2} \cos(2\pi F_0 \tau)$$

The process is **Wide-Sense Stationary (WSS)** because its mean is constant and its autocorrelation depends only on  $\tau$ .

## B.3 Power Spectral Density

The PSD  $S_Y(F)$  is the Fourier Transform of  $R_{YY}(\tau)$ :

$$S_Y(F) = \mathcal{F} \left\{ \frac{1}{2} \cos(2\pi F_0 \tau) \right\} = \frac{1}{4} [\delta(F - F_0) + \delta(F + F_0)]$$